PLEASE READ CAREFULLY

# OLFOVEL 500MG TABLET

**CONTENTS**
Each tablet contains levofloxacin hemihydrate 512.5 mg equivalent to levofloxacin 500.0 mg.

**DESCRIPTION**
Pale orange, long round biconvex film coated tablets with bisec-line and letter B, L on one side and figure 500 on the other side.

**INDICATIONS**
Treatment of adults (more than or equal to 18 years) with mild, moderate and severe infections caused by susceptible strains of the designated microorganisms in the conditions listed as follows.
Acute maxillary sinusitis due to *Streptococcus pneumoniae*, *Haemophilus influenzae* or *Moraxella catarrhalis.*
Acute bacterial exacerbation of chronic bronchitis due to *Staphylococcus aureus, Streptococcus pneumoniae, Haemophilus influenzae, Haemophilus parainfluenzae* or *Moraxella catarrhalis**.
Community-acquired pneumonia due to *Staphylococcus aureus, Streptococcus pneumoniae, Legionella pneumophila* or *Mycoplasma pneumoniae*.
Uncomplicated skin and skin structure infections (mild to moderate) including abscesses, cellulitis, furuncles, impetigo, pyoderma, wound infections, due to *Staphylococcus aureu*s or *Streptococcus pyogenes.*
Complicated urinary tract infections (mild to moderate) due to *Enterococcus faecalis, Enterobacter cloacae, Escherichia coli, Klebsiella pneumoniae, Proteus mirabilis* or *Pseudomonas aeruginosa**.
Acute pyelonephritis (mild to moderate) caused by *Escherichia col*i.
Appropriate culture and susceptibility tests should be performed before treatment in order to isolate and identify organisms causing the infection and to determine their susceptibility to levofloxacin. Therapy with levofloxacin may be initiated before results of these tests are known; once results become available, appropriate therapy should be selected. As with other drugs in this class, some strains of *Pseudomonas aeruginosa* may develop resistance fairly rapidly during treatment with levofloxacin. Culture and susceptibility testing performed periodically during therapy will provide information about the continued susceptibility of the pathogens to the antimicrobial agent and the possible emergence of bacterial resistance.
Consideration should be given to official guidance on the appropriate use of antibacterial agents.

***Olfovel 500mg Tablet** should only be used:
• When Pseudomonas is considered AND patient is allergic to antipseudomonal penicillins/cephalosporins;
• For resistant organisms with no other alternative antibiotics available.

**MODE OF ADMINISTRATION**
For oral administration.

**RECOMMENDED DOSAGE**
**Olfovel** tablets are available in the strength of 500 mg only and may not be suitable for all dosage recommendations as stated. Therefore, other suitable available strengths and/or dosage forms of levofloxacin should be used in such cases.

**Usual dose:** 250 or 500 mg administered orally, as indicated by infection and described in the following dosing chart. These recommendations apply to patients with normal renal function (i.e. creatinine clearance > 80 mL/min). For patients with altered renal function, see Table 2. Oral doses should be administered at least 2 hours before or 2 hours after antacids containing magnesium, aluminium, as well as sucralfate, metal cations, such as iron and multivitamin preparations with zinc or didanosine (Videx), chewable/buffered tablets or the paediatric powder for oral solution.

**Patient with Normal Renal Function: See Table 1**

| **Table 1** | | | | |
|---|---|---|---|---|
| **Infection*** | **Unit Dose (mg)** | **Frequency** | **Duration (Days)** | **Daily Dose (mg)** |
| Acute bacterial exacerbation of chronic bronchitis | 500 | Every 24 hours | 7 | 500 |
| Community-acquired pneumonia | 500 | Every 24 hours | 7 – 14 | 500 |
| Acute maxillary sinusitis | 500 | Every24 hours | 10 – 14 | 500 |
| Uncomplicated skin and skin structure infections | 500 | Every 4 hours | 7 – 10 | 500 |
| Complicated UTI | 250 | Every 24 hours | 10 | 250 |
| Acute pyelonephritis | 250 | Every 24 hours | 10 | 250 |
| * Due to the designated pathogen (see section **INDICATIONS**)<br>** Sequential therapy (IV to oral) may be instituted at the discretion of the physician. | | | | |

**Patient with Impaired Renal Function: See Table 2**

| **Table 2** | | |
|---|---|---|
| **Renal Status** | **Initial Dose** | **Subsequent Dose** |
| Acute bacterial exacerbation of chronic bronchitis/ community-acquired pneumonia/acute maxillary sinusitis/ uncomplicated skin and skin structure infections | | |
| $CL_{CR}$ 50 – 80 mL/min | No dosage adjustment required | |
| $CL_{CR}$ 20 – 49 mL/min | 500 mg | 250 mg every 24 hours |
| $CL_{CR}$ 10 – 19 mL/min | 500 mg | 250 mg every 48 hours |
| Haemodialysis | 500 mg | 250mg every 48 hours |
| CAPD | 500 mg | 250 mg every 48 hours |
| Complicated UTI/acute pyelonephritis. | | |
| $CL_{CR}$ ≥ 20 mL/min | No dosage adjustment required | |
| $CL_{CR}$ 10 – 19 mL/min | 250 mg | 250 mg every 48 hours |

$CL_{CR}$ = Creatinine Clearance
CAPD= Chronic ambulatory peritoneal dialysis

When only the serum creatinine is known, the following formula may be used to estimate creatinine clearance.

Males:

Creatinine Clearance (mL/min) =

$$\frac{\text{Weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females:
0.85 x value calculated using formula for males.

The serum creatinine should represent a steady state of renal function.

**CONTRAINDICATIONS**
Persons with a history of hypersensitivity to levofloxacin, quinolone antimicrobial agents or any other components of **Olfovel 500mg Tablet.**

**WARNINGS AND PRECAUTIONS**
**Warnings**
Fluoroquinolones, including levofloxacin are associated with an increased risk of tendonitis and tendon rupture in all ages. The risk of developing fluoroquinolone-associated tendinitis and tendon rupture is further increased in older patients usually over 60 years of age, in patients taking corticosteroid drugs, and in patients with kidney, heart or lung transplants. Patient experiencing pain, swelling, inflammation or a tendon or tendon rupture should be advised to stop taking levofloxacin and to contact their health care professional promptly about changing their antimicrobial therapy. Patients should also avoid exercise and using the affected area at the first sign of tendon pain, swelling or inflammation. Tendon rupture can occur during or after completion of therapy; cases occurring up to several months after completion of therapy have been reported.

Exacerbation of myasthenia gravis
Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post-marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolone use in persons with myasthenia gravis. Avoid levofloxacin in patients with known history of myasthenia gravis.

- The safety and efficacy of levofloxacin in children, adolescents (< 18 years), pregnant and nursing women have not been established.
- Convulsions and toxic psychoses have been reported in patients receiving quinolones, including levofloxacin. Quinolones may also cause increased intracranial pressure and central nervous system stimulation which may lead to tremors, restlessness, anxiety, suicidal thoughts, or acts. These reactions may occur following the 1st dose. If these reactions occur in patients receiving levofloxacin, the drug should be discontinued and appropriate measures instituted.
- As with other quinolones, levofloxacin should be used with caution in patients with a known or suspected CNS disorder that may predispose to seizures or lower than seizure threshold (e.g. severe cerebral arteriosclerosis, epilepsy) or in the presence of other risk factors that may predispose to seizures or lower than seizure threshold (e.g. certain drug therapy, renal dysfunction).
- Serious and occasionally fatal hypersensitivity and/or anaphylactic reactions have been reported in patients receiving therapy with quinolones including levofloxacin. These reactions often occur following the 1st dose. Some reactions have been accompanied by cardiovascular collapse, hypotension/shock, seizure, loss of consciousness, tingling, angioedema (including tongue, laryngeal, throat, or facial oedema/swelling), airway obstruction (including bronchospasm, shortness of breath and acute respiratory distress), dyspnoea, urticarial, itching, and other skin reactions. Levofloxacin should be discontinued immediately at first appearance of a skin rash or any other signs of hypersensitivity. Serious acute hypersensitivity reactions may require treatment with epinephrine and other resuscitative measures, including oxygen, i.v. fluids, antihistamines, corticosteroids, pressor amine, and airway management, as clinically indicated.
- Serious and sometimes fatal events, some due to hypersensitivity, and some due to uncertain aetiology, have been reported rarely in patients receiving therapy with quinolones including levofloxacin. These events may be severe and generally occur following the administration of multiple doses. Clinical manifestations may include one or more of the following: fever, rash or severe dermatologic reactions (e.g. toxic epidermal necrolysis, Stevens-Johnson syndrome); vasculitis; arthralgia, myalgia; serum sickness; allergic pneumonitis; interstitial nephritis; acute renal insufficiency or failure; hepatitis; jaundice; acute hepatic necrosis or failure; anaemia, including haemolytic and aplastic; thrombocytopenia, including thrombotic thrombocytopenic purpura; leukopenia; agranulocytosis; pancytopenia; and/or other haematologic abnormalities. **Olfovel 500mg Tablet** should be discontinued immediately at the first appearance of a skin rash or any other sign of hypersensitivity and supportive measures instituted.
- Pseudomembranous colitis has been reported with nearly all antibacterial agents, including levofloxacin, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of any antibacterial agent.
- Treatment with antibacterial agents alters the normal flora of the colon and may permit overgrowth of clostridia. Studies indicate that a toxin produced by *Clostridium difficile* is on primary cause of 'antibiotic-associated colitis'.
- After diagnosis of pseudomembranous colitis has been established, therapeutic measures should be initiated. Mild cases of pseudomembranous colitis usually respond to drug discontinuation alone. In moderate to severe cases, consideration should be given to management with fluids and electrolytes, protein supplementation and treatment with an antibacterial drug clinically effective against *C. difficile colitis.*
- The use of levofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section **ADVERSE EFFECTS**). Treatment of these patients with levofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit-risk assessment.

Aortic aneurysm and dissection
Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of

other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).
In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Prolonged, disabling and potentially irreversible serious adverse drug reactions
Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Levofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Tendinitis and tendon rupture
Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral may occur as early as within 48hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with levofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilization).
Corticosteroids should not be used if signs of tendinopathy occur.

Peripheral neuropathy
Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with levofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section **ADVERSE EFFECTS**).

Psychiatric reactions
Psychiatric reactions may occur even after the first administration of fluoroquinolones, including **Olfovel 500mg Tablet**. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section **ADVERSE EFFECTS**). In the event that the patient develops these reactions, **Olfovel 500mg Tablet** should be discontinued and appropriate measures instituted. Caution is recommended if **Olfovel 500mg Tablet** is to be used in psychotic patients or in patients with a history of psychiatric disease.

**Precautions**

- Although levofloxacin is more soluble than other quinolones, adequate hydration of patients receiving levofloxacin should be maintained to prevent the formation of a highly concentrated urine.
- Administer levofloxacin with caution in the presence of renal insufficiency. Careful clinical observation and appropriate laboratory studies should be performed prior to and during therapy since elimination of levofloxacin may be reduced. In patients with impaired renal function (creatinine clearance < 50 mL/min), adjustment of the dosage regimen is necessary to avoid the accumulation of levofloxacin due to decreased clearance.
- Moderate to severe phototoxicity reactions have been observed in patients exposed to direct sunlight while receiving drugs in this class. Excessive exposure to sunlight should be avoided. Therapy should be discontinued if phototoxicity (e.g. skin eruption) occurs.
- As with other quinolones, levofloxacin should be used with caution in any patients with a known or suspected CNS disorder that may predispose to seizures or lower the seizure threshold (e.g. severe cerebral arteriosclerosis, epilepsy) or in the presence of other risk factors that may predispose to seizures or lower the seizure threshold (e.g. certain drug therapy, renal dysfunction).
- As with other quinolones, disturbances of blood glucose, including symptomatic hyper and hypoglycaemia, have been reported, usually in diabetic patients receiving concomitant treatment with an oral hypoglycaemic agent (e.g. glyburide/glibenclamide) or with insulin. In these patients, careful monitoring of blood glucose is recommended. If a hypoglycaemic reaction occurs in a patient being treated with levofloxacin, levofloxacin should be discontinued immediately and appropriate therapy should be initiated immediately.
- Some quinolone, including levofloxacin, have been associated with prolongation of the QT interval on the electrocardiogram and infrequent cases of arrhythmia. During post-marketing surveillance, very rare cases of Torsades de Pointes have been reported in patients taking levofloxacin. These reports generally involved patients with concurrent medical conditions or concomitant medications that may have been contributory. The risk of arrhythmias may be reduced by avoiding concurrent use with other drugs that prolong the QT interval including Class IA or Class III antiarrhythmic agents; in addition, use of levofloxacin in the presence of risk factors for Torsades de Pointes, such as hypokalaemia, significant bradycardia and cardiomyopathy should be avoided.
- As with any potent antimicrobial drug, periodic assessment of organ system functions including renal, hepatic, and haematopoietic, is advisable during therapy.

**Information for Patients:**
Patients should be advised: to drink fluids liberally; those antacids containing magnesium or aluminium, as well as sucralfate, metal cations, e.g. iron, and multivitamin preparations with zinc or didanosine (Videx) chewable/buffered tablets or the paediatric powder for oral solution should be taken at least 2 hours before or 2 hours after oral levofloxacin administration; that oral levofloxacin can be taken without regard to meals; to discontinue treatment and inform the physician if the patient experiences pain, inflammation or rupture of a tendon, and to rest and refrain from exercise until the diagnosis of tendinitis or tendon rupture has been confidently excluded; that levofloxacin may be associated with hypersensitivity reactions, even following the 1st dose, and to discontinue the drug at the first sign of a skin rash, hives or other skin reactions, a rapid heartbeat, difficulty in swallowing or breathing, any swelling suggesting angioedema (e.g., swelling of the lips, tongue, face, tightness of the throat, hoarseness), or other symptoms of an allergic reaction; to avoid excessive sunlight or artificial ultraviolet light while receiving levofloxacin and to discontinue therapy if phototoxicity (i.e., skin eruption) occurs; that if diabetic and being treated with insulin or an oral hypoglycaemic agent and a hypoglycaemic reaction occurs, levofloxacin should be discontinued and consult a physician; that concurrent administration of warfarin and levofloxacin has been associated with increases of the International Normalized Ratio (INR) or prothrombin time and clinical episodes of bleeding. Patients should notify their physician if they are taking warfarin; that convulsions have been reported in patients taking quinolones, including levofloxacin and to notify their physician before taking **Olfovel 500mg Tablet** if there is a history of this condition.

**INTERACTIONS WITH OTHER MEDICAMENTS**

Effect of other medicinal products on **Olfovel 500mg Tablet:**
*Iron salts, magnesium- or aluminium-containing antacids, didanosine*
Levofloxacin absorption is significantly reduced when iron salts, or magnesium- or aluminium- containing antacids, or didanosine *(only didanosine formulations with aluminium or magnesium containing buffering agents)* are administered concomitantly with **Olfovel 500mg Tablet.** Concurrent administration of fluoroquinolones with multi-vitamins containing zinc appears to reduce their oral absorption. It is recommended that preparations containing divalent or trivalent cations such as iron salts, zinc salts or magnesium- or aluminium-containing antacids or didanosine (only didanosine formulations with aluminium or magnesium containing buffering agents) should not be taken 2 hours before or after **Olfovel 500mg Tablet** administration.

Calcium salts have a minimal effect on the oral absorption of levofloxacin.

*Sucralfate*
The bioavailability of levofloxacin is significantly reduced when administered together with sucralfate. If the patient is to receive both sucralfate and levofloxacin, it is best to administer sucralfate 2 hours after **Olfovel 500mg Tablet** administration.

*Theophylline, fenbufen or similar non-steroidal anti-inflammatory drugs*
No pharmacokinetic interactions of levofloxacin were found with theophylline. However, a pronounced lowering of the cerebral seizure threshold may occur when quinolones are given concurrently with theophylline, nonsteroidal anti-inflammatory drugs, or other agents which lower the seizure threshold.
Levofloxacin concentrations were about 13% higher in the presence of fenbufen than when administered alone.

*Probenecid and cimetidine*
Probenecid and cimetidine had a statistically significant effect on the elimination of levofloxacin. The renal clearance of levofloxacin was reduced by cimetidine (24%) and probenecid (34%). This is because both drugs are capable of blocking the renal tubular secretion of levofloxacin.

Caution should be exercised when levofloxacin is co-administered with drugs that effect the tubular renal secretion such as probenecid and cimetidine, especially in renally impaired patients.

Effect of **Olfovel 500mg Tablet** on other medicinal products
*Ciclosporin*
The half-life of ciclosprorin was increased by 33% when co-administered with levofloxacin.

*Vitamin K antagonists*
Increased coagulation tests (PT/INR) and/or bleeding, which may be severe, have been reported in patients treated with levofloxacin in combination with a vitamin K antagonist (e.g. warfarin). Coagulation tests, therefore, should be monitored in patients treated with vitamin K antagonists.

*Drugs known to prolong QT interval*
Levofloxacin, like other fluoroquinolones, should be used with caution in patients receiving drugs known to prolong the QT interval (e.g. Class IA and III antiarrhythmics, tricyclic antidepressants, macrolides, antipsychotics).

Other forms of interactions
*Food*
**Olfovel 500mg Tablet** may be administered regardless of food intake.

**PREGNANCY AND LACTATION**
Pregnancy
There are, however, no adequate and well-controlled studies in pregnant women. Levofloxacin should be used during pregnancy only if the potential benefit justifies potential risk to the foetus.

Lactation
Levofloxacin has not been measured in human milk. Based upon data from ofloxacin, it can be presumed that levofloxacin will be excreted in human milk. Because of the potential for serious adverse reactions from levofloxacin will be excreted in human milk. Because of the potential for serious adverse reactions from levofloxacin in nursing infants, a decision should be made whether to discontinue nursing or to discontinue the drug taking into account the importance of the drug to the mother.

**ADVERSE EFFECTS**

| System organ Class | Common | Uncommon | Rare | Very rare | Not known |
|---|---|---|---|---|---|
| Infections and infestations | | Fungal infection Including candida infection<br>Pathogen resistance | | | |
| Blood and lymphatic system disorder | | Leukopenia<br>Eosinophilia | Thrombocytopenia<br>Neutropenia | | Pancytopenia<br>Haemolytic anaemia<br>Agranulocytosis |
| Immune system disorders | | | Angioedema<br>Hypersensitivity | | Anaphylactic[a] shock<br>Anaphylactoid[a] shock |
| Metabolism and nutrition disorders | | Anorexia | Hypoglycaemia particularly in diabetic patients | | Hyperglycaemia<br>Hypoglycaemic coma |
| Psychiatric disorders* | Insomnia | Anxiety<br>Confusional state | Psychotic reactions (with e.g. | Psychotic reactions (potentially | Psychotic disorders with self-endangering |

| | | | | | |
|---|---|---|---|---|---|
| | | Nervousness | hallucination, paranoia)<br>Depression<br>Agitation<br>Abnormal dreams<br>Nightmares | culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide) | behaviour including suicidal ideation or suicide attempt |
| Nervous system disorders* | Headache<br>Dizziness | Somnolence<br>Tremor<br>Dysgeusia | Paraesthesia<br>Convulsion | | Peripheral sensory neuropathy<br>Peripheral sensory motor neuropathy<br>Parosmia including anosmia<br>Dyskinesia<br>Extrapyramidal disorder<br>Ageusia<br>Syncope<br>Benign intracranial hypertension |
| Eye disorders* | | | Visual disturbances such as blurred vision | | Transient vision loss |
| Ear and labyrinth disorders* | | Vertigo | Tinnitus | | Hearing loss<br>Hearing impaired |
| Cardiac disorders** | | | Tachycardia,<br>Palpitation | | Ventricular tachycardia,<br>which may result in cardiac arrest<br>Ventricular arrhythmia and Torsades de Pointes (reported predominantly in patients with risk factors of QT prolongation),<br>electrocardiogram QT prolonged |
| Vascular disorders** | | | Hypotension | | |
| Respiratory, thoracic and mediastinal disorders | | Dyspnoea | | | Bronchospasm<br>Pneumonitis allergic |
| Gastrointestinal disorders | Diarrhoea<br>Vomiting<br>Nausea | Abdominal pain<br>Dyspepsia<br>Flatulence<br>Constipation | | | Pancreatitis<br>Diarrhoea – haemorrhagic<br>which in very rare cases may be indicative of enterocolitis, including pseudomembranous colitis |
| Hepatobiliary disorders | Hepatic enzyme Increased (ALT/AST, Alkaline phosphatase, GGT) | Blood bilirubin increased | | | Jaundice and severe liver injury, including cases with fatal acute liver failure, primarily in patients with severe underlying diseases<br>Hepatitis |
| Skin and subcutaneous tissue disorders[b] | | Rash<br>Pruritus<br>Urticaria<br>Hyperhidrosis | | | Toxic epidermal necrolysis<br>Stevens-Johnson syndrome<br>Erythema multiforme<br>Photosensitivity reaction<br>Leukocytoclastic vasculitis<br>Stomatitis |

| | | | | | |
|---|---|---|---|---|---|
| Musculoskeletal and connective tissue disorders* | | Arthralgia<br>Myalgia | Tendon disorders including tendinitis (e.g. Achilles tendon)<br>Muscular weakness which may be of special importance in patients with myasthenia gravis | | Rhabdomyolysis<br>Tendon rupture (e.g. Achilles tendon)<br>Ligament rupture<br>Muscle rupture<br>Arthritis |
| Renal and urinary disorders | | Blood creatinine increased | Renal failure acute (e.g. due to interstitial nephritis) | | |
| General disorders and administration site conditions* | | Asthenia | Pyrexia | | Pain (including pain in back, chest, and extremities) |

[a] Anaphylactic and anaphylactoid reactions may sometimes occur even after the first dose.
[b] Mucocutaneous reactions may sometimes occur even after the first dose.
Other undesirable effects which have been associated with fluoroquinolone administration include:
- attacks of porphyria in patients with porphyria.

* Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendonitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of quinolones and fluoroquinolones in some cases irrespective of pre-existing risk factors (see section **WARNINGS AND PRECAUTIONS**).
** Cases of aortic aneurysm and dissection, sometimes complicated by rupture (including fatal ones), and of regurgitation/incompetence of any of the heart valves have been reported in patients receiving fluoroquinolones.

<u>Post Marketing Experience</u>
Exacerbation of myasthenia gravis

**SYMPTOMS & TREATMENT OF OVERDOSAGE**
Levofloxacin exhibits a low potential for acute toxicity. In the event of an acute overdosage, the stomach should be emptied. The patient should be observed and appropriate hydration maintained. Levofloxacin is not efficiently removed by haemodialysis or peritoneal dialysis.

**EFFECTS ON ABILITY TO DRIVE AND USE MACHINES**
Certain undesirable effects (e.g. dizziness/vertigo, drowsiness, visual disturbances) may impair the patient's ability to concentrate and react, and therefore may constitute a risk in situations where these abilities are of special importance (e.g. driving a car or operating machinery).

**PHARMACODYNAMICS**
Pharmacotherapeutic group: Quinolone antibacterials – Fluoroquinolone
ATC code: J01MA12
Levofloxacin is a synthetic antibacterial agent of the fluoroquinolone class and is the S (-) enantiomer of the racemic drug substance ofloxacin.

**Mechanism of action**
As a fluoroquinolone antibacterial agent, levofloxacin acts on the DNA-DNA-gyrase complex and topoisomerase IV.

**PK/PD relationship**
The degree of the bactericidal activity of levofloxacin depends on the ratio of the maximum concentration in serum (Cmax) or the area under the curve (AUC) and the minimal inhibitory concentration (MIC).

**Mechanism of resistance**
Resistance to levofloxacin is acquired through a stepwise process by target site mutations in both type II topoisomerases, DNA gyrase and topoisomerase IV. Other resistance mechanisms such as permeation barriers (common in *Pseudomonas aeruginosa*) and efflux mechanisms may also affect susceptibility to levofloxacin.

Cross-resistance between levofloxacin and other fluoroquinolones is observed. Due to the mechanism of action, there is generally no cross-resistance between levofloxacin and other classes of antibacterial agents.

**Antibacterial spectrum**
The prevalence of resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

**<u>Commonly susceptible species</u>**
**Aerobic Gram-positive bacteria**
*Bacillus anthracis*
*Staphylococcus aureus methi-S*
*Staphylococcus saprophyticus*
*Streptococci, group C and G*
*Streptococcus agalactiae*
*Streptococcus pneumoniae*
*Streptococcus pyogenes*

**Aerobic Gram- negative bacteria**
*Eikenella corrodens*
*Haemophilus influenzae*
*Haemophilus parainfluenzae*
*Klebsiella oxytoca*
*Moraxella catarrhalis*
*Pasteurella multocida*
*Proteus vulgaris*
*Providencia rettgeri*

**Anaerobic bacteria**
*Peptostreptococcus*

**Other**
*Chlamydophila pneumoniae*
*Chlamydophila psittaci*
*Chlamydia trachomatis*
*Legionella pneumophila*
*Mycoplasma pneumoniae*
*Mycoplasma hominis*
*Ureaplasma urealyticum*

**Species for which acquired resistance may be a problem**
**Aerobic Gram-positive bacteria**
*Enterococcus faecalis*
*Methicillin-Resistant Staphylococcus aureus (MRSA)*#
*Coagulase-Negative Staphylococcus spp*

**Aerobic Gram-negative bacteria**
*Acinetobacter baumannii*
*Citrobacter freundii*
*Enterobacter aerogenes*
*Klebsiella pneumoniae*
*Enterobacter cloacae*
*Escherichia coli*
*Morganella morganii*
*Proteus mirabilis*
*Providencia stuartii*
*Pseudomonas aeruginosa*
*Serratia marcescens*

**Anaerobic bacteria**
*Bacteroides fragilis*

**Inherently Resistant Strains**
**Aerobic Gram-positive bacteria**
*Enterococcus faecium*

# MRSA are very likely to possess co-resistance to fluoroquinolones.

**PHARMACOKINETICS**
**Absorption**
Orally administered levofloxacin is rapidly and almost completely absorbed with peak plasma concentrations being obtained within 1 to 2 hours. The absolute bioavailability is approximately 99% to 100%.

Food has little effect on the absorption of levofloxacin.

Steady state conditions are reached within 48 hours following a 500 mg once or twice daily dosage regimen.

**Distribution**
Approximately 30% to 40% of levofloxacin is bound to serum protein.
The mean volume of distribution of levofloxacin is approximately 100 L after single and repeated 500 mg doses, indicating widespread distribution into body tissues.

**Penetration into tissues and body fluids:**
Levofloxacin has been shown to penetrate bronchial mucosa, epithelial lining fluid, alveolar macrophages, lung tissue, skin (blister fluid), prostatic tissue and urine. However, levofloxacin has poor penetration intro cerebrospinal fluid.

**Biotransformation**
Levofloxacin is metabolised to a very small extent, the metabolites being desmethyl-levofloxacin and levofloxacin N-oxide. These metabolites account for < 5% of the dose excreted in urine. Levofloxacin is stereo-chemically stable and does not undergo chiral inversion.

**Elimination**
Following oral and intravenous administration of levofloxacin, it is eliminated relatively slowly from the plasma ($t_{1/2}$: 6 to 8 hours). Excretion is primarily by the renal route > 85 % of the administered dose.

The mean apparent total body clearance of levofloxacin following a 500 mg single dose was 175 ± 29.2 mL/min.

There are no major differences in the pharmacokinetics of levofloxacin following intravenous and oral administration, suggesting that the oral and intravenous routes are interchangeable.

**Linearity**
Levofloxacin obeys linear pharmacokinetics over a range of 50 to 1000 mg.

**Special populations**
**Patients with renal impairment**
The pharmacokinetics of levofloxacin are affected by renal impairment. With decreasing renal function renal elimination and clearance are decreased, and elimination half-lives increased as shown in the table below:
Pharmacokinetics in renal insufficiency following single oral 500 mg dose

| $CL_{CR}$ (ml/min) | < 20 | 20 – 49 | 50 – 80 |
|---|---|---|---|
| $CL_{CR}$ (ml/min) | 13 | 26 | 57 |
| $t_{1/2}$ (h) | 35 | 27 | 9 |

**Elderly patients**
There are no significant differences in levofloxacin kinetics between young and elderly patients, except those associated with differences in creatinine clearance.

**Gender differences**
Separate analysis for male and female patients showed small to marginal gender differences in levofloxacin pharmacokinetics. There is no evidence that these gender differences are of clinical relevance.

**STORAGE CONDITIONS**
Keep in tight container. Protect from light.
Do not store above 30 °C.
Keep out of reach of children.
*Jauhi daripada kanak-kanak.*

**SHELF LIFE**
Please refer to the outer box label.

**PACKING/PACK SIZES**
Aluminium-PVDC blister strip of 10 tablets per strip. Box of 3x10, 5x10, 10x10, 30x10 and 50x10 tablets.

**MANUFACTURED BY**
T.O. CHEMICALS (1979) LTD.
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